

SHIWEN (LAREINA) YANG

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EDUCATION

Cornell Tech (Cornell University)

Master of Engineering in Computer Science | GPA: 3.8/4.0

Aug 2024 – May 2025

New York, US

Dalhousie University

Bachelor of Computer Science: Honours Co-op in Computer Science | GPA: 4.12/4.3

Sep 2019 – May 2024

Halifax, NS

Honors: First Class Honours, Women in Technology Scholarship, Sexton Scholar

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, SQL, HTML, CSS, PHP, Swift, C, C#, R

Frameworks/Tools: ReactJS, Redux, Tailwind, NodeJS, NextJS, REST, GraphQL, Jest, Storybook, SpringBoot, Docker, AWS

Databases: MySQL, PostgreSQL, MongoDB, Firebase (Firestore)

EXPERIENCE

ReelData AI

Sep 2022 – Dec 2023

Software Developer

Halifax, NS

- Developed the camera presets feature end-to-end, designing the database schema (Flyway), building RESTful APIs (Java/Spring Boot), and leveraged GraphQL/OpenAPI CodeGen for type-safe API integration in React, improving development efficiency and feature reliability
- Improved dashboard usability by implementing a dynamic sidebar and responsive layout in React, enhancing navigation and data visualization
- Increased code reliability by writing Storybook-based UI tests integrated into the CI/CD pipeline
- Resolved UI freezes by profiling the application and refactoring components to eliminate main-thread blocking operations
- Collaborated cross-functionally with product managers and UI/UX designers throughout the SDLC on scalable architecture, code reviews, and deployment

Collab (Dear Life)

Jan 2022 – Apr 2022

Full Stack Developer

Halifax, NS

- Developed the onboarding flow—including sign-in/sign-up, verification, feature tour, and password setup—for a Flutter-based mobile app designed to help users record life stories
- Integrated AWS Cognito for authentication, Aurora MySQL for serverless data management, Mixpanel for user interaction analytics, and Sentry for real-time error tracking
- Designed and evaluated three user flow prototypes, improving usability and engagement
- Supported early product discovery by conducting pain point interviews and designing user research questionnaires

PINet Lab

May 2021 – Aug 2021

Research Assistant

Halifax, NS

- Modified the JavaScript-based TSCH-Sim simulator to model three network environments—with custom node topologies and radio link formations—and tested network parameters to optimize PHY layer performance across setups
- Analyzed literature, synthesized textual materials and employed Python to generate analytical charts from experimental data
- Contributed to a GLOBECOM paper, *BlueTiSCH: A Multi-PHY Simulation of Low-Power 6TiSCH IoT Networks*

PROJECTS

TaskMosaic, (React, Next.js, Postgres, Drizzle, BetterAuth, Tailwind, Gemini API, AWS)

Spring 2025

- A Next.js full-stack web app that auto-generates and assigns subtasks based on project descriptions and team member skills
- Implemented a real-time dashboard with drag-and-drop and backend sync styled with shaden/ui and Tailwind CSS
- Deployed using Supabase, Docker, and AWS ECS/EC2 with a full CI/CD pipeline

AI Art Detector, (Python, Scikit-learn, Streamlit)

Spring 2025

- A machine learning system to distinguish between Stable Diffusion-generated and human-created Ukiyo-e artworks
- Implemented two models: a logistic regression model using engineered features (entropy, line count, saturation) with 99.8% accuracy and strong interpretability; and a CNN trained on raw images, achieving 98.3% accuracy
- Deployed an interactive Streamlit app with a clean two-tab UI guiding users from image upload to model prediction

Search Term Identification for Concept Location Leveraging Word Relations, (Java, Python)

Spring 2023, Spring 2024

- Enhance query reformulation technique, STRICT using text graphs with three types of word relations to improve concept search accuracy
- Investigated combinations of five graph-based algorithms and their weight distribution for search term selection
- Proposed a novel algorithm—STRICT++ that integrates four graph-based algorithms with three types of inter-word relations
- Conducted experiments on 946 change requests from 22 systems, demonstrating performance improvements of STRICT++